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National Plan Report United States – 2022 National Security Strategy (NSS)

Evaluated through Classical Economic Planning Typologies

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Executive Summary

This report evaluates the 2022 Biden-Harris National Security Strategy (NSS) as the United States' closest equivalent to a national development plan. Using the classical typologies of economic planning as outlined in the course framework, the analysis systematically assesses the NSS across twelve dimensions, including inducement versus direction, democratic versus totalitarian, centralized versus decentralized, functional versus structural, and general versus partial planning.

The findings demonstrate that the NSS reflects a comprehensive inducement-based and democratic planning model, mobilizing fiscal incentives, public-private partnerships, and legislative frameworks to guide national priorities. It embodies general planning by integrating agriculture, industry, services, infrastructure, environment, and governance under a unified strategic vision. The NSS also exhibits flexibility and national scope, applying uniformly across all states and regions while promoting equity through initiatives such as Justice40 and regional technology hubs.

At the same time, the evaluation identifies critical gaps. The NSS lacks a formal long-term perspective plan spanning 15–25 years, instead operating as a strategic doctrine with annual cycles. Physical resource allocation is less emphasized compared to financial inducements, and while the plan is adaptive, it does not fully institutionalize rolling frameworks that anticipate unforeseen shocks. These limitations highlight the need for stronger integration of perspective planning, physical resource mobilization, and structural transformation mechanisms.

Overall, the NSS functions as a multidimensional national framework that reconciles market efficiency with strategic state guidance. It strengthens democratic governance, secures supply chains, advances climate resilience, and promotes balanced regional growth. By situating the NSS within the typological framework of economic planning, this report underscores its role as a *de facto* national plan for the United States in the 2020s, while also recommending adjustments to enhance its coherence, durability, and long-term impact.

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Chapter 1

Introduction

1.1 Background and Context

National development planning has historically been a cornerstone of economic governance, guiding countries in mobilizing resources, setting priorities, and balancing growth with equity. While the United States does not publish a formalized “national development plan” in the classical sense, the 2022 Biden-Harris National Security Strategy (NSS) functions as its closest equivalent. Issued by the White House, the NSS integrates economic, social, environmental, and governance objectives under a unified strategic vision, positioning national security as inseparable from economic resilience and sustainable development.

1.2 Purpose of the Report

This report evaluates the 2022 NSS through the lens of classical economic planning typologies taught in the Economics Development course. By applying frameworks such as inducement versus direction, democratic versus totalitarian, centralized versus decentralized, and functional versus structural planning, the analysis situates the NSS within the broader tradition of economic planning. The purpose is twofold:

- To assess how effectively the NSS serves as a de facto national plan for the United States.
- To identify strengths, weaknesses, opportunities, and threats that shape its long-term viability.

1.3 Research Questions

The evaluation is guided by the following key questions:

- To what extent does the NSS embody the characteristics of classical economic planning?
- How does the NSS reconcile market efficiency with strategic state guidance?
- What gaps exist in terms of long-term perspective, resource mobilization, and structural transformation?
- How do internal strengths and weaknesses interact with external opportunities and threats in shaping outcomes?

1.4 Methodology and Approach

The analysis employs a typological framework derived from classical economic planning theory. Twelve dimensions—including inducement versus direction, democratic versus totalitarian, and general versus partial planning—are systematically applied to the NSS. Complementary tools such as SWOT analysis and sectoral evaluation provide additional insights into internal and external factors. The methodology emphasizes both qualitative interpretation and quantitative evidence, drawing on legislative acts (CHIPS and Science Act, Infrastructure Investment and Jobs Act, Inflation Reduction Act), program evaluations (Justice40), and macroeconomic indicators.

Chapter 2

SWOT Analysis

2.1 SWOT Analysis of the 2022 National Security Strategy (NSS)

The following SWOT analysis evaluates the 2022 Biden-Harris National Security Strategy (NSS) through the lens of economic planning and strategic governance. It synthesizes internal strengths and weaknesses alongside external opportunities and threats, highlighting the NSS’s positioning within both domestic and global contexts.

Table 1: SWOT Analysis of the 2022 National Security Strategy (NSS)

Strengths	Weaknesses
Strong use of incentives – subsidies, tax credits, PPPs mobilize investment without command planning.	Lack of long-term horizon – focuses on short/medium cycles, missing 15–25 year foresight.
Democratic foundation – built on legislation, transparency, and civic participation (Justice40).	Heavy reliance on fiscal inducements – underplays constraints in land, water, minerals, industrial capacity.
Comprehensive sectoral coverage – integrates agriculture, industry, services, infrastructure, environment, governance.	Uneven implementation capacity – state/local variations limit uniform national impact.
Flexible and adaptive implementation – rolling reviews, resilience frameworks allow recalibration to crises.	Limited structural transformation – functional focus does not fully address inequality or outdated structures.
Strong global leadership orientation – alliances, climate agreements, trade partnerships reinforce U.S. influence.	Incomplete equity outcomes – Justice40 targets unmet; disadvantaged communities lack full access.
Opportunities	Threats
Global demand for clean energy & green tech – export potential for U.S. green industries.	Geopolitical instability & global shocks – conflicts and supply chain disruptions undermine strategies.
Strengthened democratic alliances – cooperation enhances supply chain resilience and climate coordination.	Macroeconomic uncertainties – inflation, recession, deficits, interest volatility constrain resources.

Rising semiconductor demand – AI, automation, cloud computing drive chip expansion.	Intensifying climate impacts – disasters strain resilience and divert funds.
Advances in emerging technologies – quantum, biotech, robotics, energy storage align with U.S. strengths.	Technological bottlenecks – shortages in labor, minerals, grid capacity delay expansion.
Growth in global climate finance & cooperation – multilateral funding expands leadership opportunities.	Political polarization & policy reversals – shifts in priorities threaten continuity.

Analysis of Internal Factors

1. Strengths

1.1. Strong use of incentives

The NSS emphasizes inducement mechanisms such as subsidies, tax credits, and public-private partnerships. This approach mobilizes private investment in strategic sectors like semiconductors, clean energy, and advanced manufacturing. By avoiding command-based planning, it maintains market dynamism while steering resources toward national priorities.

1.2. Democratic foundation and stakeholder participation

Grounded in Congressional legislation and civic engagement, the NSS ensures transparency and legitimacy in policymaking. Programs like Justice40 embed equity and public participation, giving communities a voice in implementation. This democratic anchoring strengthens accountability and broadens support for national initiatives.

1.3. Comprehensive sectoral coverage

The strategy integrates agriculture, industry, services, infrastructure, environmental protection, and governance under one framework. This holistic scope prevents fragmented policy development and ensures coherence across economic and social domains. It positions the NSS as a unifying doctrine rather than a collection of isolated programs.

1.4. Flexible and adaptive implementation

Rolling reviews and resilience frameworks allow the NSS to recalibrate in response to crises such as pandemics, supply chain disruptions, or climate shocks. This adaptability enhances national

resilience and ensures that strategic objectives remain relevant in volatile global conditions.

1.5. Strong global leadership orientation

By coordinating with allies, participating in global agreements, and leading on climate policy, the NSS reinforces U.S. influence internationally. This dual orientation — domestic resilience paired with global leadership — strengthens the country’s strategic position while aligning internal development with external commitments.

2. Weaknesses

2.1. Lack of a long-term strategic horizon

The NSS emphasizes short- to medium-term cycles rather than adopting a 15–25 year perspective. This limits foresight in areas such as infrastructure transformation, climate adaptation, and systemic industrial restructuring. Without a long-range vision, the strategy risks being reactive rather than anticipatory.

2.2. Heavy reliance on fiscal inducements

Financial incentives such as subsidies and tax credits dominate the NSS framework. While effective in mobilizing investment, this reliance underplays constraints tied to physical resources — land, water, minerals, and industrial capacity. The imbalance between financial inducements and resource planning may weaken sustainability.

2.3. Uneven implementation capacity

Variations in state and local administrative capability create inconsistencies in execution. Some regions lack the institutional strength to fully implement federal directives, resulting in uneven national impact and widening disparities in outcomes.

2.4. Limited structural transformation

The NSS’s functional sectoral focus does not fully address deeper systemic issues such as inequality, outdated industrial structures, or labor-capital imbalances. Without structural reforms, the strategy risks reinforcing existing inefficiencies rather than transforming the economic base.

2.5. Incomplete equity outcomes

Justice40, while ambitious, has yet to fully deliver benefits to disadvantaged communities. Many vulnerable groups still face barriers to accessing clean energy investment and climate resilience programs. This gap undermines the equity-inclusive vision of the NSS and weakens its legitimacy.

Analysis of External Factors

1. Opportunities

1.1. Global demand for clean energy and green technology

The accelerating worldwide transition toward renewable energy and climate-friendly technologies creates significant export potential for U.S. industries. By leveraging its innovation capacity and policy support, the U.S. can position itself as a leading supplier of clean energy solutions, enhancing both economic growth and climate leadership.

1.2. Strengthened democratic alliances

Deepening cooperation with allies bolsters supply chain resilience, technology sharing, and coordinated climate responses. These alliances not only reinforce U.S. strategic priorities but also expand collective capacity to address global challenges, from energy security to technological innovation.

1.3. Rising semiconductor demand worldwide

Global reliance on advanced chips for AI, automation, and cloud computing provides strong momentum for U.S. semiconductor expansion. This demand aligns with domestic investment strategies, offering opportunities to secure technological leadership and reduce dependence on foreign supply chains.

1.4. Advances in emerging technologies

Breakthroughs in quantum computing, biotechnology, robotics, and energy storage open new frontiers for innovation. The U.S. can capitalize on these advances by integrating them into national security and economic planning, reinforcing its competitive edge in high-tech sectors.

1.5. Growth in global climate finance and cooperation

International climate funding and multilateral initiatives create avenues for the U.S. to expand leadership and mobilize investment. By engaging in global climate finance, the NSS can amplify domestic programs while strengthening international credibility and influence.

2. Threats

2.1. Geopolitical instability and global shocks

Conflicts, great-power competition, and disrupted supply chains pose significant risks to the NSS. These shocks can undermine industrial expansion, weaken climate strategies, and divert resources toward immediate security concerns, reducing the effectiveness of long-term planning.

2.2. Macroeconomic uncertainties

Inflationary pressures, recessions, fiscal deficits, and interest-rate volatility threaten the financial stability required for sustained investment. Such macroeconomic turbulence can erode private-sector confidence, slow capital mobilization, and constrain the fiscal space available for inducement-based policies.

2.3. Intensifying climate impacts

Increasing frequency and severity of hurricanes, floods, droughts, and wildfires strain national resilience. These disasters divert funds from strategic projects to emergency relief, weakening the NSS's ability to maintain momentum in infrastructure transformation and climate adaptation.

2.4. Technological bottlenecks

Shortages of skilled labor, critical minerals, grid capacity, and specialized equipment risk delaying semiconductor, clean energy, and advanced manufacturing expansion. These bottlenecks highlight vulnerabilities in supply chains and workforce readiness that could stall strategic objectives.

2.5. Political polarization and policy reversals

Sharp shifts in federal priorities between administrations create uncertainty in climate policy, industrial strategy, and global partnerships. This instability undermines continuity, discourages

long-term investment, and weakens international confidence in U.S. commitments.

Chapter 3

Typological Evaluation

3.1 Typological Evaluation of the 2022 National Security Strategy

To visually synthesize the typological evaluation, the following table is inserted to highlight how the 2022 Biden-Harris National Security Strategy aligns with classical economic planning dimensions.

Table 2: Typological Evaluation of the 2022 NSS

Typology Dimension	NSS Classification	Justification
Inducement vs. Direction	Inducement-based	Relies on fiscal incentives, subsidies, and public–private partnerships rather than direct command.
Democratic vs. Totalitarian	Democratic	Anchored in legislative acts, transparency, and civic participation (Justice40, stakeholder engagement).
Centralized vs. Decentralized	Decentralized	Federal vision cascades through state and local governments with implementation flexibility.
Functional vs. Structural	Functional	Focuses on sectoral functions (energy, infrastructure, human capital) rather than rigid structural quotas.
General vs. Partial	General	Integrates agriculture, industry, services, infrastructure, environment, and governance under one framework.
Flexible vs. Rigid	Flexible	Adaptive mechanisms, rolling reviews, and resilience planning allow recalibration in response to shocks.

National vs. Regional	National scope	Applies uniformly across all states, with equity mechanisms to balance regional disparities.
Perspective vs. Annual	Annual cycles	Operates as a strategic doctrine with short-term cycles; lacks a formal 15–25 year perspective plan.
Financial vs. Physical	Financial emphasis	Strong reliance on inducements and subsidies; limited integration of physical resource allocation.
Comprehensive vs. Sectoral	Comprehensive	Covers multiple sectors simultaneously, ensuring coherence across economic, social, and environmental domains.
Domestic vs. Global	Dual orientation	Balances domestic resilience with global leadership in climate, trade, and security.
Equity vs. Efficiency focus	Equity-inclusive	Justice40 and regional hubs embed fairness while pursuing efficiency gains in productivity and competitiveness.

Source: Author’s evaluation based on NSS 2022

1. **Direction vs. Inducement Planning**

The NSS is firmly inducement-based. Rather than mandating production quotas or directing output, it mobilizes investment through fiscal incentives such as subsidies, tax credits, and public-private partnerships. Legislative acts like the CHIPS and Science Act, Infrastructure Investment and Jobs Act (IIJA), and Inflation Reduction Act (IRA) exemplify inducement planning by steering private capital into strategic sectors. This approach maintains market dynamism while aligning investment with national priorities, making the NSS a textbook case of inducement planning.

2. **Democratic vs. Totalitarian Planning**

The NSS is anchored in democratic institutions and participatory governance. It derives legitimacy from Congressional legislation, transparency mechanisms, and civic engagement. Programs such as Justice40 require agencies to consult disadvantaged communities, embedding equity and accountability into implementation. This participatory

orientation ensures that planning is rights-preserving and inclusive, contrasting sharply with totalitarian models that rely on coercion or centralized command.

3. Centralized vs. Decentralized Planning

The NSS demonstrates centralized strategic coherence through federal legislation and White House direction, while decentralizing execution to states, local governments, and regional hubs. This hybrid model ensures national priorities remain unified while allowing responsiveness to diverse regional contexts.

4. Functional vs. Structural Planning

Although the NSS emphasizes functional priorities such as energy transition, infrastructure modernization, and human capital development, its structural dimension remains partial. Reshoring industries and climate transition commitments represent initial structural reforms, but the NSS does not yet institutionalize comprehensive systemic transformation across inequality, industrial restructuring, or labor-capital balances.

5. General vs. Partial Planning

The NSS embodies general planning by integrating agriculture, industry, services, infrastructure, environment, and governance under one framework. This holistic scope prevents fragmentation and ensures coherence across economic and social domains. By covering multiple sectors simultaneously, the NSS functions as a unifying doctrine rather than a collection of isolated programs.

6. Perspective vs. Annual Planning

The NSS operates primarily on annual cycles, with reviews and budget appropriations guiding short-term action. Yet it embeds perspective elements through long-term commitments such as achieving net-zero emissions by 2050 and sustaining global technological leadership. This duality reflects a partial but evolving orientation toward perspective planning.

7. Rolling vs. Fixed Planning

While not formally codified as rolling plans, the NSS incorporates resilience frameworks and adaptive reviews that function in practice as rolling adjustments. These mechanisms enable recalibration in response to shocks

such as pandemics, supply chain disruptions, or climate extremes, aligning the NSS more closely with rolling than fixed planning.

8. Physical vs. Financial Planning

The NSS relies heavily on financial inducements—subsidies, tax credits, and public-private partnerships—yet physical resource mobilization remains underdeveloped. Workforce readiness, mineral supply, land use, and grid infrastructure capacity are critical constraints that require stronger integration. A balanced approach would ensure financial incentives are matched with physical resource planning to sustain long-term resilience.

9. Indicative vs. Imperative Planning

The NSS is clearly indicative, providing guidance and incentives rather than imposing mandatory directives. It signals national priorities through inducements and legislation, leaving room for private sector flexibility. This indicative orientation aligns with democratic governance and market efficiency.

10. Socialistic vs. Capitalistic Planning

The NSS reflects a capitalist-democratic model, relying on market mechanisms and private sector participation. Inducements encourage entrepreneurship and innovation, while equity mechanisms such as Justice40 embed fairness. This balance positions the NSS within capitalist planning traditions, distinct from socialistic command models.

11. Flexible vs. Rigid Planning

The NSS exemplifies flexible planning by embedding adaptive mechanisms, rolling reviews, and resilience frameworks, while deliberately avoiding rigid quotas or centralized output mandates. This flexibility ensures responsiveness to evolving economic and security conditions, distinguishing the NSS from rigid planning models.

12. Regional vs. National vs. International Planning

The NSS operates at multiple scales. At the national level, it applies uniformly across all states. At the regional level, mechanisms such as Justice40 and technology hubs promote equity and balanced growth. At the international level, alliances, climate agreements, and trade partnerships reinforce U.S. leadership. This multi-scale orientation ensures coherence between domestic resilience and global responsibilities.

Chapter 4

Strategic Planning Framework

The 2022 Biden-Harris National Security Strategy (NSS) serves as the overarching strategic framework guiding U.S. national development priorities throughout the current decade. While not legally binding in the manner of a formal economic plan, the NSS functions as a strategic umbrella that integrates legislative acts, executive programs, and interagency coordination into a coherent national vision. Its role is to provide direction, coherence, and resilience across economic, social, environmental, and governance domains.

1. Institutional Anchoring

The NSS is issued by the White House and endorsed across federal agencies, ensuring that its priorities cascade through the Department of Commerce, Department of Energy, Department of Defense, Department of Transportation, and other key institutions. This anchoring provides legitimacy and durability, aligning national security objectives with economic development imperatives.

2. Legislative Integration

The NSS is operationalized through major legislative instruments that embody its strategic intent:

- **CHIPS and Science Act (2022–2027):** Catalyzes domestic semiconductor manufacturing and advanced technology innovation.
- **Infrastructure - Investment and Jobs Act (2022–2026):** Modernizes transportation, energy grids, and broadband infrastructure.
- **Inflation Reduction Act (2022–2030):** Mobilizes clean energy investment and climate resilience measures.

Together, these acts translate the NSS vision into actionable programs, ensuring fiscal mobilization and sectoral coverage.

3. Cross-Sectoral Coordination

The NSS emphasizes a whole-of-government approach, requiring coordination across federal, state, and local levels. This framework ensures:

- **Economic Security:** Strengthening supply chains and revitalizing domestic manufacturing.

- **Human Capital Development:** Expanding education, workforce training, and healthcare innovation.
- **Environmental Sustainability:** Advancing net-zero emissions goals and climate-resilient infrastructure.
- **Governance Capacity:** Reinforcing democratic institutions, transparency, and civic engagement.

4. **Strategic Coherence**

By integrating national security with economic development, the NSS establishes coherence between domestic priorities and international commitments. It positions the United States as a global leader in climate action, technological innovation, and democratic governance, while ensuring that national resilience is embedded in every sector.

Chapter 5

Climate and Energy Transition

Climate action is positioned at the core of the 2022 Biden-Harris National Security Strategy (NSS), reflecting the recognition that environmental sustainability is inseparable from national security and economic resilience. The framework integrates climate goals with energy policy, infrastructure modernization, and global leadership commitments, making the transition to clean energy both a domestic priority and an international responsibility.

To illustrate the current status of Justice40 implementation, the following table summarizes program-level evaluations across 445 federally covered initiatives. This visual is based on the January 2024 report by Walls, Hines, and Ruggles, which tracks agency progress toward meeting the 40% equity goal. The table categorizes programs by implementation status, highlighting both the scale of federal engagement and the challenges in achieving full compliance.

Table 3: Justice40 Implementation Status Across 445 Federal Programs

Status Number	Number of Programs	Percentage of Programs Evaluated
U (Uncategorized)	42	9%
1 (No information available)	133	30%
2 (Acknowledged but no detail)	40	9%
3 (Preliminary guidance only)	52	12%
4 (Defined metrics, no 40% guarantee)	80	18%
5A (Confirmed 40% benefit delivery)	10	2%
5B (Automatically qualifies: tribal/low-income)	88	20%
Total Programs Evaluated	445	100%

Source: Walls, Hines, and Ruggles (2024), Justice40 Implementation Status Report

1. Net-Zero Commitments

The NSS reaffirms U.S. commitments to achieve net-zero greenhouse gas emissions by 2050, with interim targets of a 50–52 percent reduction by 2030 relative to 2005 levels. These goals are embedded in legislative instruments such as the Inflation Reduction Act, which mobilizes large-scale investment in renewable energy, electrification, and carbon capture technologies.

2. Clean Energy Innovation

The strategy emphasizes innovation as the driver of the energy transition. Federal inducements—tax credits, subsidies, and procurement contracts—are designed to catalyze private sector investment in:

- **Renewable energy technologies** (solar, wind, geothermal, and offshore projects).
- **Next-generation energy storage** (batteries, hydrogen, and advanced grid systems).
- **Carbon management solutions** (carbon capture, utilization, and storage).
- **Nuclear energy modernization** as a reliable low-carbon option.

This innovation agenda positions the United States as a global leader in clean technology development.

3. Climate-Resilient Infrastructure

The NSS links climate action to infrastructure resilience. The Infrastructure Investment and Jobs Act provides funding for modernizing energy grids, expanding broadband, and strengthening transportation systems against climate-related disruptions. Investments in flood defenses, wildfire mitigation, and drought resilience are integrated into broader national security planning.

4. Regional Equity in Transition

Recognizing disparities across regions, the NSS incorporates equity mechanisms such as the Justice40 Initiative, which directs 40 percent of climate and clean energy benefits to disadvantaged communities. This ensures that the energy transition promotes inclusive growth, reduces regional vulnerabilities, and strengthens social cohesion.

5. Global Climate Leadership

The NSS situates U.S. climate policy within a global framework. By recommitting to the Paris Agreement and advancing international climate finance, the United States seeks to reinforce alliances, support developing countries, and maintain credibility as a leader in global climate governance. This outward orientation underscores the dual role of climate policy as both domestic transformation and international diplomacy.

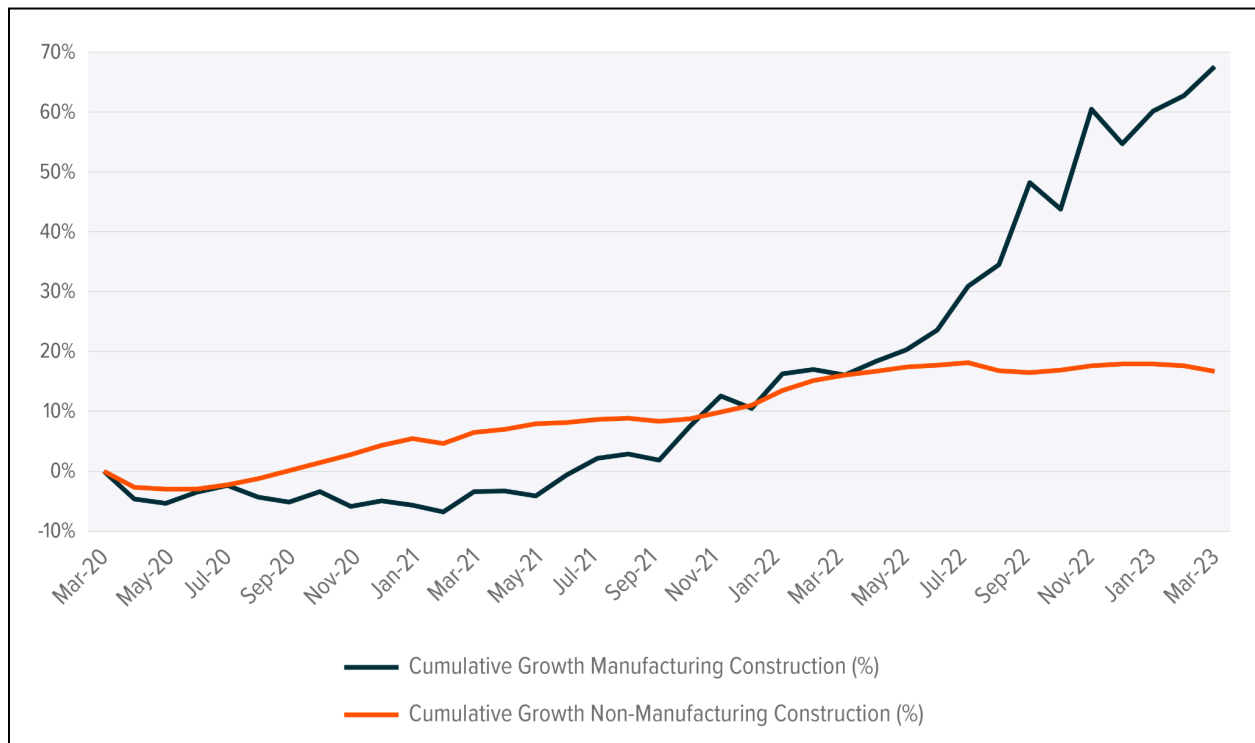
Chapter 6

Fiscal and Economic Outlook

The 2022 Biden-Harris National Security Strategy (NSS) frames economic strength as foundational to national security, aligning fiscal mobilization with strategic investment across technology, infrastructure, energy, and human capital. While the NSS itself does not present macro-fiscal tables, its operationalization through major legislation—including the Inflation Reduction Act (IRA), CHIPS and Science Act, and Infrastructure Investment and Jobs Act (IIJA)—anchors a multi-year public investment agenda intended to crowd in private capital, raise productivity, and harden resilience against shocks.

Recent data from the U.S. Treasury and Census Bureau confirm that this investment-led strategy is already reshaping the economic landscape. The following figures illustrate key trends in manufacturing construction, business fixed investment, and R&D growth—all of which validate the NSS’s fiscal stance and long-term sustainability logic.

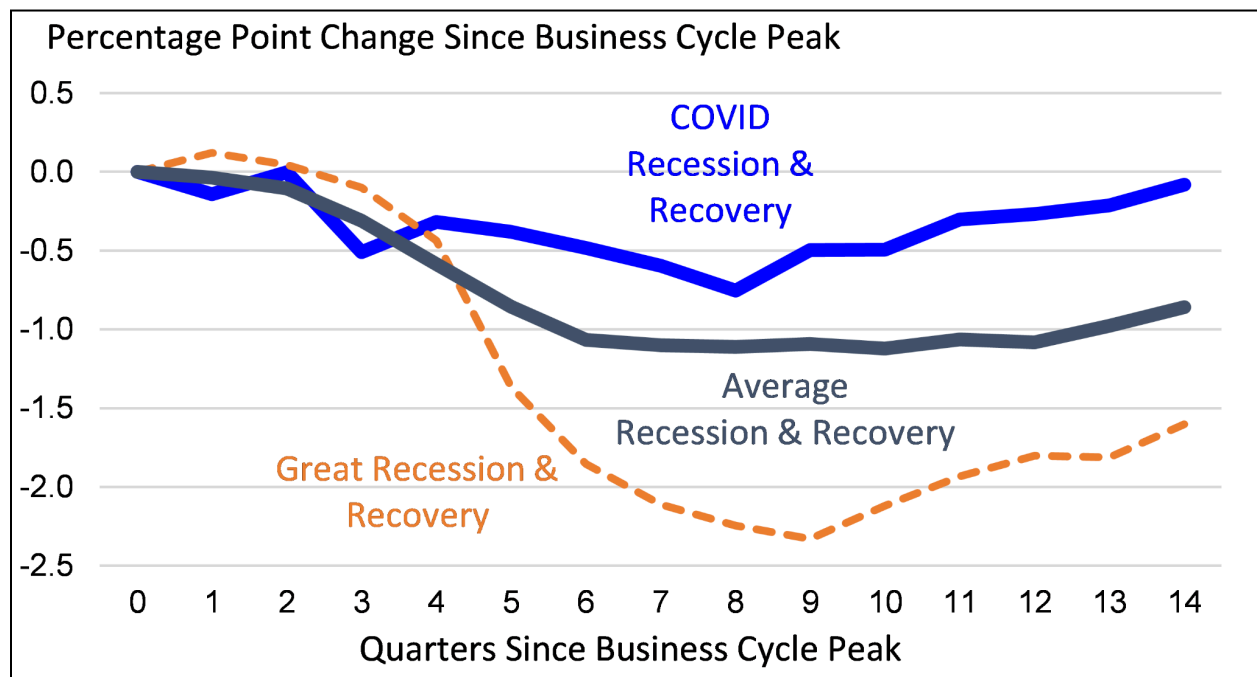
Figure 1: Cumulative Growth in U.S. Manufacturing vs. Non-Manufacturing Construction Spending, Mar 2020–Mar 2023



Source: Bloomberg, L.P. (n.d.). News Trends. Accessed on May 1, 2023

This figure shows that manufacturing-related construction spending surged nearly 70% since the pandemic began, compared to only ~10% growth in non-manufacturing sectors. The CHIPS Act and IRA are major drivers of this divergence, reflecting targeted inducements for industrial capacity expansion.

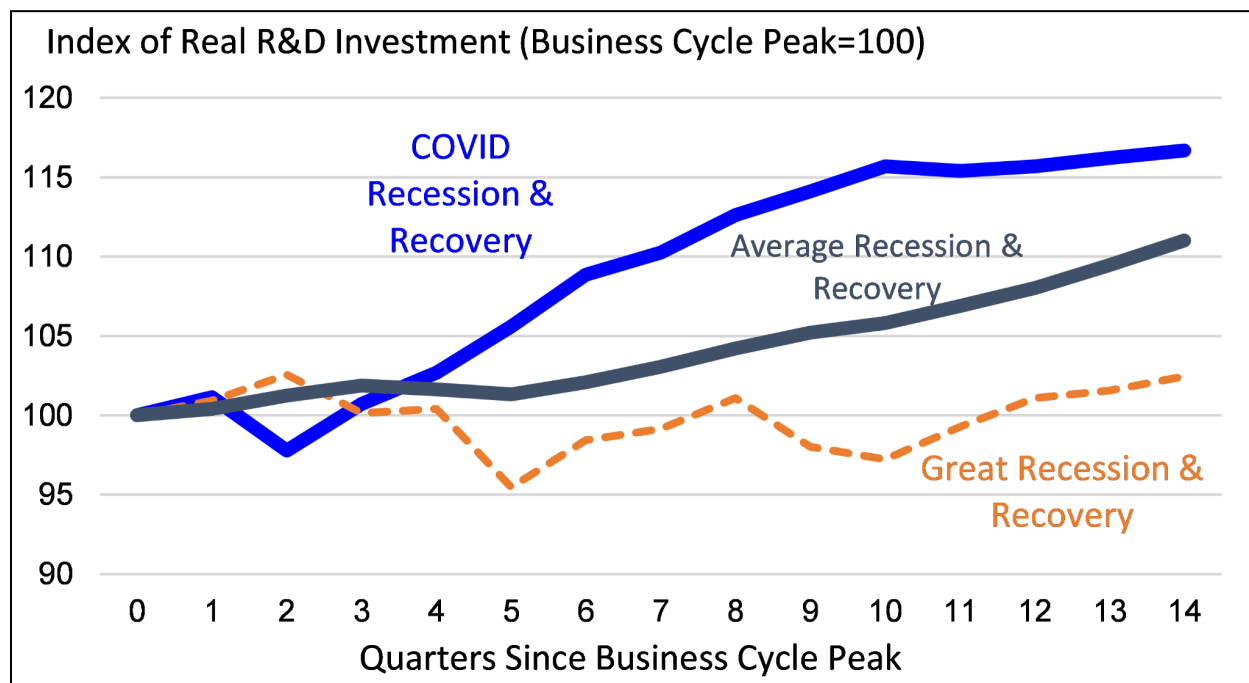
Figure 2: Business Fixed Investment as Share of GDP – COVID vs. Past Recessions



Source: Bureau of Economic Analysis; U.S. Treasury calculations.

Unlike previous recessions, business fixed investment (BFI) has remained stable as a share of GDP during the COVID recovery. This resilience supports the NSS's claim that multi-year investment programs can sustain growth even amid macroeconomic uncertainty.

Figure 3: Private R&D Investment Index by Business Cycle



Source: Bureau of Economic Analysis; U.S. Treasury calculations

R&D investment has grown 17% post-COVID, outperforming both the Great Recession and historical averages. This trend reinforces the NSS's emphasis on innovation ecosystems and long-term productivity gains.

1. Investment-Led Growth Strategy

- Strategic federal programs channel substantial funding through tax credits, grants, and procurement to accelerate clean energy deployment, rebuild infrastructure, and reshore critical industries.
- Inducements are designed to leverage private co-investment, deepen supply chains, and expand advanced manufacturing capacity.
- Investments target bottlenecks—grid capacity, broadband, logistics, and semiconductor fabrication—to reduce structural frictions and improve total factor productivity.

2. Fiscal Stance and Sustainability

- Program authorizations span several years, aligning capital formation with policy continuity and enabling phased delivery.

- Annual appropriations and tax expenditure design embed strategic priorities into the fiscal framework without prescribing centralized output quotas.
- The approach balances near-term deficits against expected gains in capacity and resilience, with adaptation mechanisms to recalibrate pace and composition.

3. Labor Market and Human Capital

- Skills formation is emphasized via apprenticeships, community college partnerships, and STEM pipelines aligned to semiconductors, clean energy, and advanced manufacturing.
- Equity instruments broaden labor force participation and support just transition pathways.
- Public R&D and targeted incentives complement private innovation, supporting diffusion of technologies into production and services.

4. 4.4 Inflation, Supply-Side Capacity, and Resilience

- By expanding domestic capacity in energy, technology, and logistics, the NSS seeks to alleviate supply constraints that amplify price pressures.
- Clean energy scaling and grid modernization are intended to dampen volatility from fossil price cycles over time.
- Strengthened stockpiles, diversified suppliers, and regional manufacturing ecosystems reduce exposure to global disruptions, supporting macro stability.

5. External Balance and Competitiveness

- The strategy prioritizes secure, trusted supply chains, export controls in sensitive technologies, and ally-shoring to protect competitiveness.
- Tech hubs and regional innovation engines aim to distribute competitive capacity beyond traditional clusters, enhancing nationwide productivity and export potential.
- The combined agenda targets durable advantages in semiconductors, clean energy, and frontier technologies, reinforcing the U.S. growth model and external balance over the medium term.

6. Implementation Risks and Mitigations

- Delivery depends on permitting efficiency, interagency coordination, and state/local implementation quality; continuous monitoring and adaptive program design are integral.

- Financial inducements must be matched with labor, materials, land, water, and energy availability to prevent cost overruns and delays.
- Multi-year investment signals support private commitment; clear guardrails and periodic reviews mitigate uncertainty from electoral cycles and evolving global conditions.

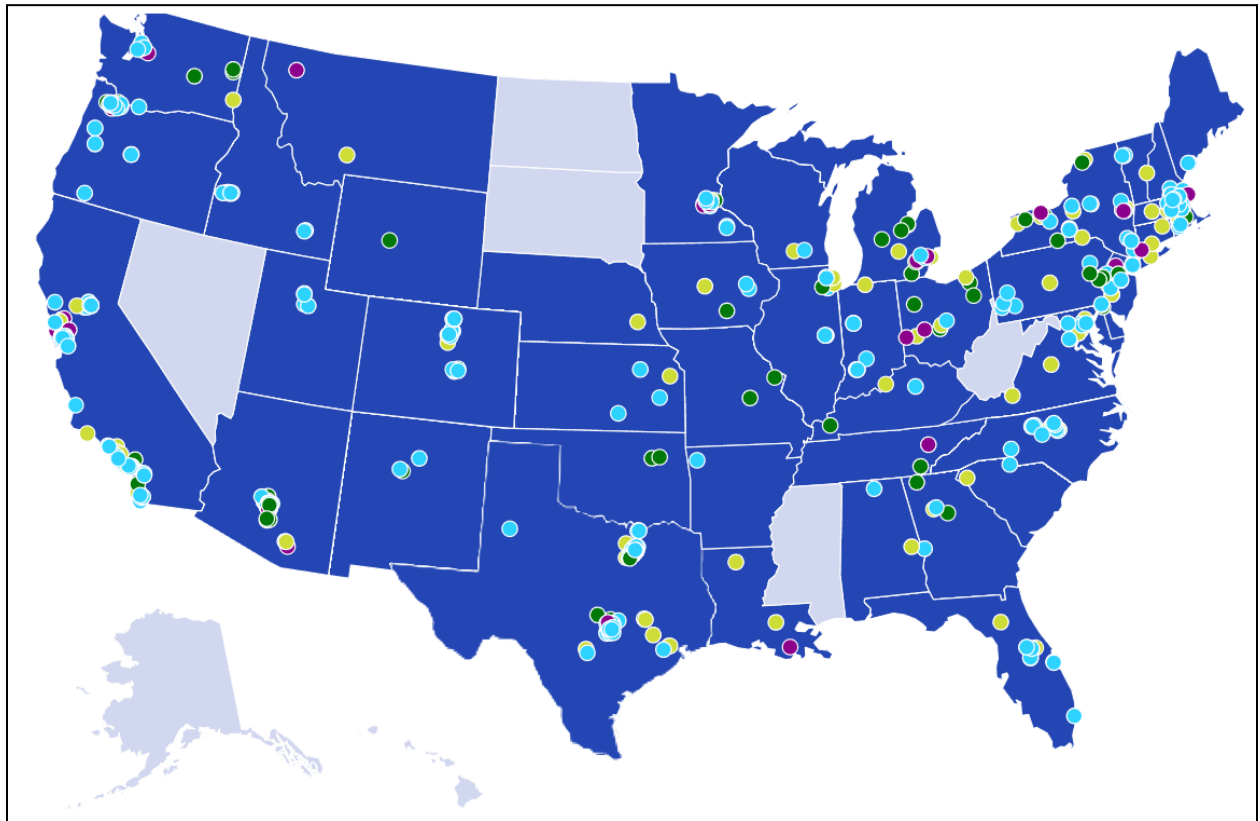
6.1. Sectoral Priorities

The 2022 Biden-Harris National Security Strategy (NSS) identifies sectoral transformation as a cornerstone of U.S. resilience, competitiveness, and inclusive growth. Its implementation spans advanced manufacturing, clean energy, digital infrastructure, and environmental sustainability—each supported by targeted inducements, regulatory reform, and public–private coordination. This section synthesizes sectoral priorities through the lens of strategic investment, institutional capacity, and typological alignment.

Agriculture and Industry

The NSS supports agricultural modernization through climate-smart practices and supply chain resilience. In industry, the CHIPS Act anchors a manufacturing revival, restoring domestic semiconductor capacity and reducing strategic dependencies.

Figure 4: U.S. Semiconductor Ecosystem Map (2025)



Source: Semiconductor Industry Association

This map visualizes the distribution of semiconductor facilities across the United States, including foundries, OSATs, equipment suppliers, materials providers, and university R&D partners—reinforcing the NSS’s commitment to industrial sovereignty and regional equity. Each color-coded dot represents a distinct segment of the ecosystem:

- Light blue for Semiconductors
- Purple for Equipment
- Dark green for Materials
- Yellow-green for University R&D Partners

The geographic spread of these facilities illustrates the CHIPS Act’s role in decentralizing industrial capacity and embedding innovation across diverse regions.

Services and Human Capital

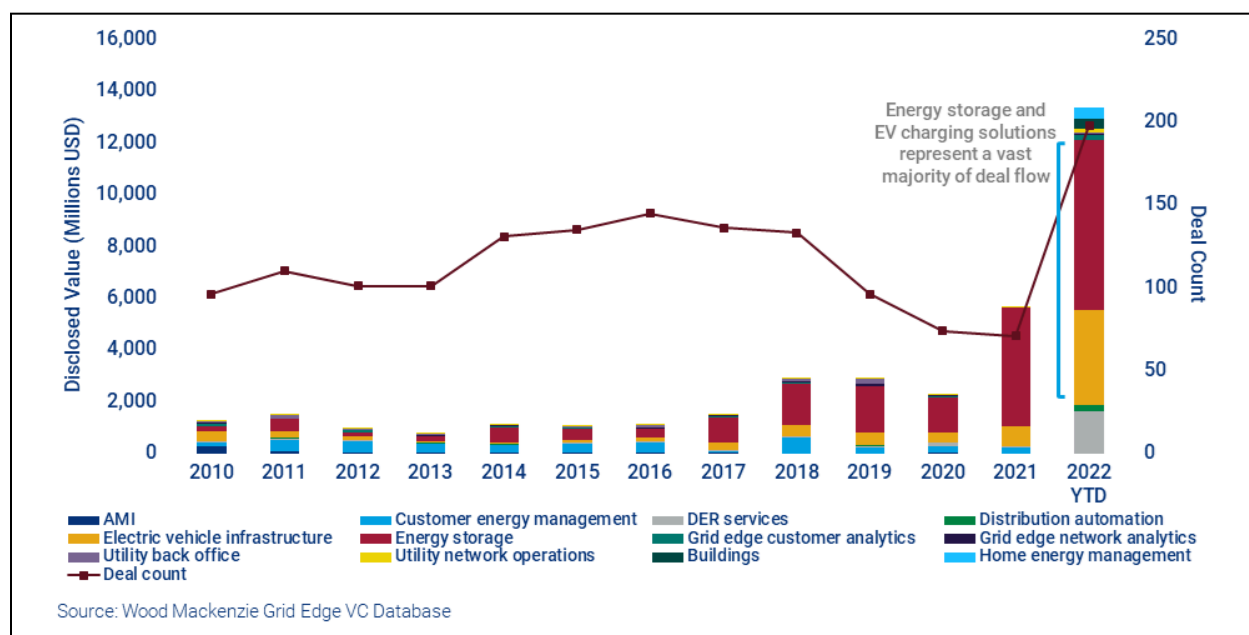
Federal programs under the IRA and CHIPS Act are expanding service-sector demand in clean energy, EV infrastructure, and digital logistics. Human capital development is embedded through workforce training, STEM pipelines, and community college partnerships.

The rise of home energy management systems, supported by IRA incentives and grid-edge innovation, reflects a shift toward service-based resilience. Partnerships between solar, storage, and automation firms (e.g., Sunrun, Enphase) signal convergence between tech services and household energy autonomy.

Infrastructure

Infrastructure investment has surged across transmission, distribution, and digital systems. Federal inducements and regulatory reforms have catalyzed both public and private capital flows.

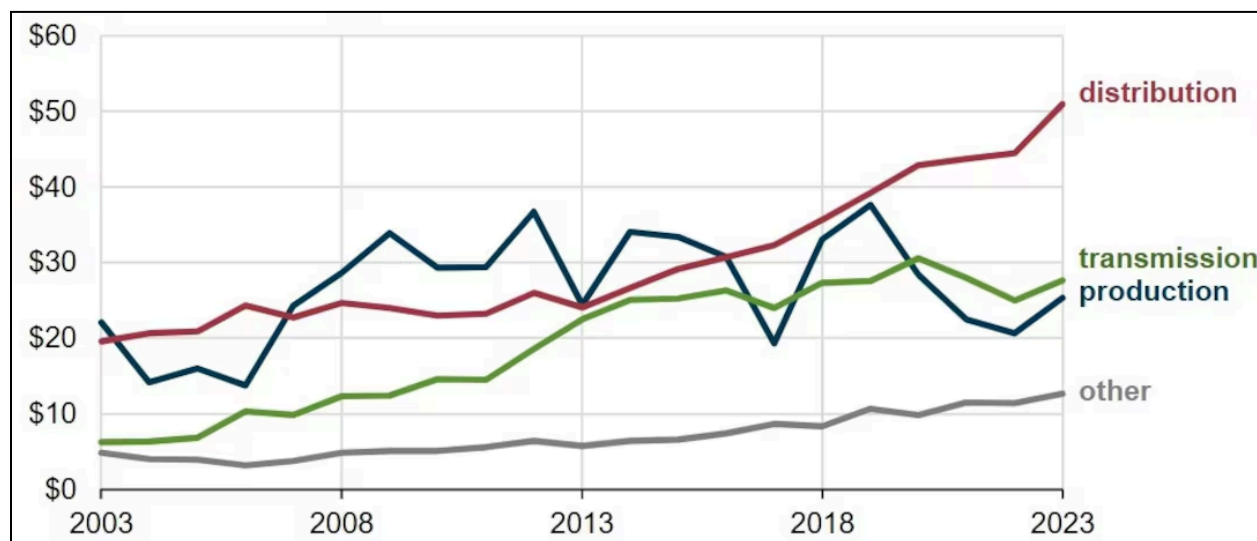
Figure 5: U.S. Grid Edge VC Investments by Market and Deal Count (2010–2021)



Source: Wood Mackenzie Grid Edge VC Database

This figure shows rising private investment in EV charging, energy storage, and analytics—validating the NSS’s inducement strategy.

Figure 6: Annual U.S. Capital Additions by Sector (2003–2023), bn dollars



Source: U.S. Energy Information Administration (EIA)

Long-term trends confirm sustained growth in transmission and distribution investment, supporting NSS goals of modernization and resilience.

Insights from Reuters highlight transmission bottlenecks, with over 2.5 TW of clean energy projects in grid queues and AI-driven data center demand projected to double by 2028. FERC Order 1920 and state-level permitting reforms are accelerating project delivery.

Environment

Environmental priorities are embedded in the NSS through clean energy scaling, emissions reduction, and climate resilience. IRA incentives target low-carbon technologies, while grid enhancements (e.g., STATCOM, advanced conductors) improve integration of intermittent renewables.

Heat pump adoption, supported by building codes and subsidies, reflects the NSS’s push toward electrification. However, cost pressures and regional disparities in breakeven thresholds require adaptive implementation.

Governance

Governance reforms under the NSS emphasize interagency coordination, stakeholder engagement, and typological flexibility. Federal inducements are matched by state-level innovations, such as Massachusetts' one-stop permitting law and regional transmission planning mandates.

The NSS's governance model is typologically functional, flexible, and decentralized, allowing sector-specific adaptation while preserving national coherence. Public-private partnerships and equity safeguards (e.g., Justice40) ensure inclusive delivery across sectors.

6.2. Governance and Institutional Capacity

The 2022 Biden-Harris National Security Strategy (NSS) reinforces governance as the backbone of national planning, emphasizing democratic institutions, transparency, and intergovernmental coordination. Institutional capacity is framed not only as a safeguard for national security but also as a driver of economic development and social resilience.

Democratic Governance

The NSS reflects a shift toward participatory governance, emphasizing transparency, stakeholder engagement, and equity safeguards. Justice40 exemplifies this through its requirement that agencies consult disadvantaged communities during program design and benefit methodology development. The White House Environmental Justice Advisory Council (WHEJAC) further recommends embedding Justice40 into agency strategic plans, creating advisory boards, and funding local organizations to increase civic participation.

Table 4: Justice40 Timeline

Date	Milestone
January 2021	Executive Order 14008 issued: 40% of benefits from federal investments to flow to disadvantaged communities
July 2021	Interim guidance released; agencies delegated to define covered programs and benefit methodologies
December 2021	Agencies submit methodologies to OMB for calculating and tracking Justice40 benefits
February 2022	Beta version of CEJST (Climate and Economic Justice Screening Tool) released for public comment
November 2022	Final version of CEJST released
January 2023	Addendum issued: agencies must fully transition to CEJST by October 1, 2023
April 2023	Initial list of 468 Justice40-covered programs released
November 2023	Updated list expands to 518 programs, including additions from the Inflation Reduction Act

Source: Resources for the Future (2024)

This timeline captures key governance milestones, including the rollout of the CEJST screening tool and deadlines for agency transition. It demonstrates how democratic governance is operationalized through phased implementation and public consultation.

Federal-State-Local Partnerships

Justice40's delivery model relies on multi-level coordination. Federal agencies define program scope and benefit metrics, while state and local governments execute spending through grants and infrastructure projects. The decentralized structure allows flexibility but introduces variation in capacity and outcomes.

Table 5: Number of Justice40 Covered Programs by Agency

Agency	Covered Programs (Apr 2023)	Programs Reviewed
Department of Energy (DOE)	146	146
Environmental Protection Agency (EPA)	73	73
U.S. Department of Agriculture (USDA)	65	65
Department of the Interior (DOI)	65	58
Department of Transportation (DOT)	39	39
Department of Housing and Urban Development (HUD)	24	24
Department of Health and Human Services (HHS)	13	13
Army Corps of Engineers (ACE)	11	11
Department of Commerce (DOC)*	9	7
Department of Homeland Security (DHS): FEMA	4	4
National Science Foundation (NSF)	3	3
National Aeronautics and Space Administration (NASA)	2	2
Other	14	0
Total	468	445

Note All 7 DOC programs reviewed are within NOAA.

Source: Resources for the Future (2024)

This figure shows how program responsibility is distributed across federal agencies, with DOE and EPA leading in volume. It underscores the importance of vertical partnerships and tailored implementation strategies.

Institutional Integration

Justice40 integrates across 518 programs spanning climate, energy, housing, transportation, and workforce development. Agencies are tasked with aligning internal processes, assigning Justice40 leads, and coordinating with the interagency EJ task force. However, integration remains uneven, with some programs lacking public documentation or clear prioritization metrics.

Table 6: Types of Federal Grants-in-Aid Programs

Type of Grant	Degree of Federal Control	Justice40 Covered Program Examples
Block Grant	Low	HUD Community Development Block Grant, HHS Community Services Block Grant, DOT FHWA Transportation Alternatives Set-Aside in Surface Transportation Block Grant Program
Project Categorical	High	FEMA Building Resilient Infrastructure and Communities Program, FEMA Flood Mitigation Assistance Grants, DOT FHWA Charging and Fueling Infrastructure Discretionary Grants
Formula Categorical	Low, within program guidelines	EPA Drinking Water and Clean Water State Revolving Loan Funds, National Park Service Land and Water Conservation Fund Stateside Grants, DOT FHWA NEVI Formula Program
Open Reimbursement	None	No Justice40 programs (example: DOT Federal-Aid Highway Program)

Source: Resources for the Future (2024)

This typology illustrates how program design affects institutional control and integration. Block grants offer flexibility but limited federal oversight, while project categorical grants enable targeted delivery with higher administrative control.

Resilience and Adaptability

The 2022 Biden-Harris National Security Strategy (NSS) governance model is typologically flexible, allowing agencies to recalibrate benefit definitions, update program lists, and respond to stakeholder feedback. Justice40's evolving scope—from 463 to 518 programs—reflects this adaptability. Agencies are encouraged to use CEJST data, refine methodologies, and transition toward outcome-based evaluation.

Justice40's challenges—such as data limitations, uneven capacity, and definitional ambiguity—are being addressed through technical assistance programs, improved tracking systems, and iterative guidance. These mechanisms enhance institutional resilience and align with the NSS's long-term governance vision.

6.3 Public Engagement and Communication

The 2022 Biden-Harris National Security Strategy (NSS) emphasizes inclusive governance and civic participation as essential components of national resilience. While the document does not prescribe detailed mechanisms for public engagement, its principles of transparency, equity, and shared values imply a commitment to participatory development and broad communication.

1. Transparency and Accountability

- **Public reporting:** Major legislative programs such as the CHIPS Act, Infrastructure Investment and Jobs Act, and Inflation Reduction Act include requirements for public disclosure of funding allocations and outcomes.
- **Oversight mechanisms:** Congressional hearings, inspector general audits, and independent evaluations ensure accountability and provide citizens with visibility into program performance.
- **Open data initiatives:** Federal agencies increasingly publish datasets on infrastructure, energy, and climate investments, enabling civil society and academia to monitor progress.

2. Civic Participation

- **Community engagement:** Federal programs encourage consultation with local communities, tribal nations, and advocacy groups to ensure that national priorities reflect diverse perspectives.
- **Equity initiatives:** The Justice40 framework embeds civic participation by directing benefits to disadvantaged communities, reinforcing trust and inclusivity.
- **Stakeholder collaboration:** Partnerships with labor unions, industry associations, and environmental organizations broaden the scope of participation in policy design and implementation.

3. Communication Strategies

- **Narrative framing:** The NSS positions democratic governance and resilience as shared national values, creating a unifying narrative for public communication.
- **Regional outreach:** Federal agencies tailor communication to regional contexts, highlighting local benefits of national programs to strengthen public support.
- **Digital platforms:** Online portals and social media channels disseminate information widely, ensuring accessibility and transparency across demographics.

4. Challenges and Opportunities

- **Limited detail:** The NSS provides broad principles but lacks explicit frameworks for sustained public engagement.
- **Opportunity for expansion:** Embedding participatory budgeting, citizen advisory councils, and regional forums could strengthen communication and deepen legitimacy.
- **Resilience through trust:** Effective communication enhances public trust, which is critical for sustaining long-term national strategies across electoral cycles.

6.4. Risks and Uncertainties

The 2022 Biden-Harris National Security Strategy (NSS) acknowledges that strategic planning must account for dynamic risks across geopolitical, economic, and environmental domains. This section synthesizes key uncertainties and outlines mitigation strategies embedded in federal programs and institutional reforms.

Geopolitical Risks

Geopolitical tensions—including trade disputes, regulatory shifts, and supply chain realignments—pose significant threats to procurement and infrastructure continuity. According to the 2024 Gartner Chief Procurement Officer Survey, 32% of procurement leaders identified geopolitical trends as a top risk to future success.

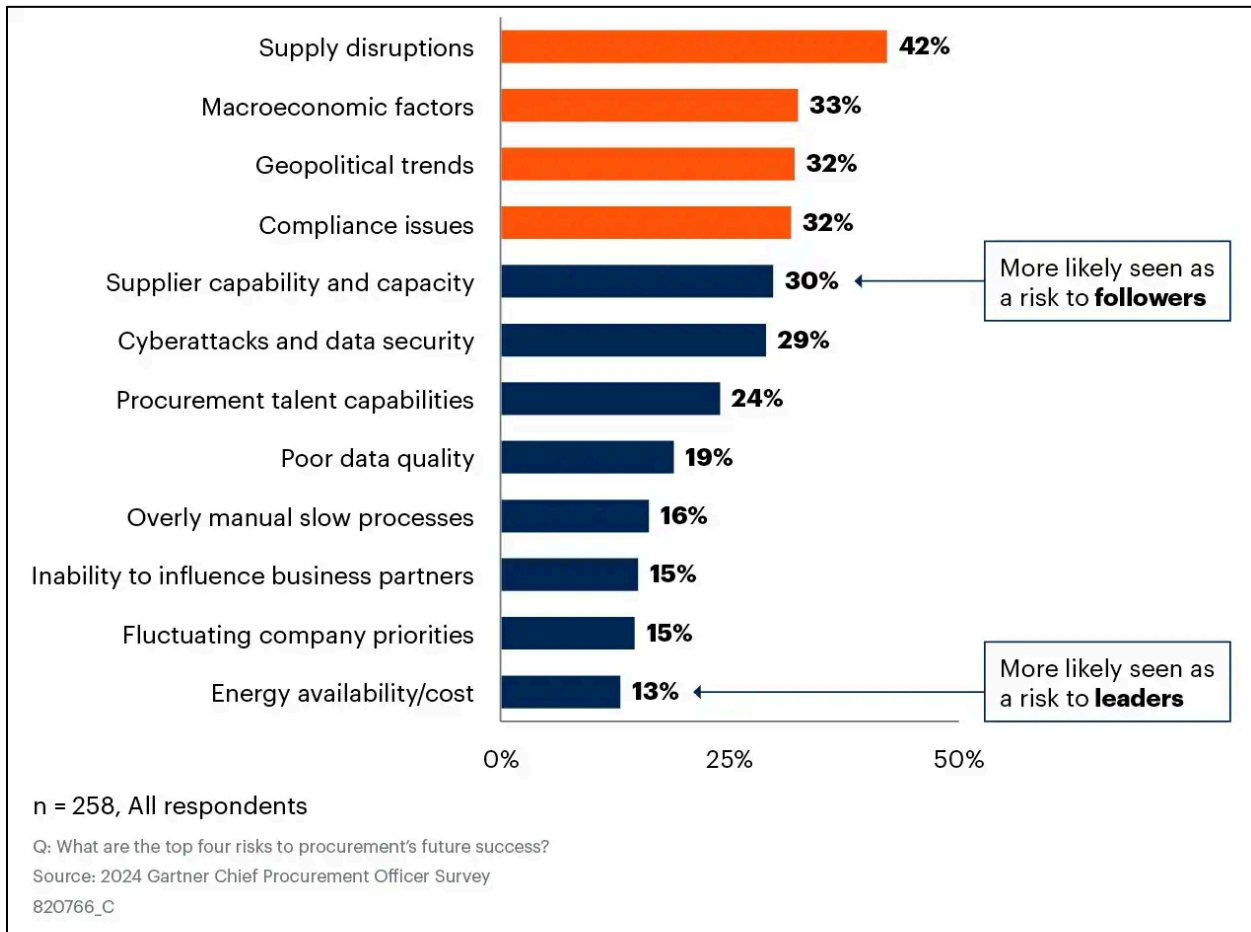
The NSS mitigates these risks through:

- Ally-shoring and export controls in sensitive technologies
- Reshoring of critical industries (e.g., semiconductors, pharmaceuticals)
- Diversification of supply chains to reduce exposure to single-region dependencies

Economic Risks

Macroeconomic instability—including inflation, interest rate volatility, and energy cost fluctuations—can undermine investment planning and delivery timelines. The same Gartner survey ranks macroeconomic factors as the second-highest procurement risk (33%).

Figure 10: Top Risks to Procurement’s Future Success (Gartner Survey, 2024)



Source: Gartner Chief Procurement Officer Survey, 2024

This bar chart presents the percentage of procurement leaders identifying key risks to future success. Supply disruptions (42%) top the list, followed by macroeconomic factors (33%), geopolitical trends (32%), and compliance issues (32%). The data reflects growing concern over unpredictable shocks and systemic vulnerabilities in global sourcing and infrastructure planning.

Additionally, grid infrastructure faces pressure from rising AI-driven electricity demand, forecasted to double by 2028, and over 2.5 TW of clean energy projects remain in grid connection queues

Environmental Risks

Environmental risks pose systemic threats to infrastructure, health, and economic stability. The NSS identifies climate resilience as a national priority, with federal programs targeting mitigation across energy, housing, transportation, and land use. However, exposure remains uneven across geographies and community types.

Recent data from Four Twenty Seven, a climate risk analytics firm, reveals stark disparities in physical climate hazards across U.S. counties. Their projections through 2040 highlight five major risks: sea level rise, hurricanes, extreme rainfall, water stress, and heat stress. To preserve the strategic focus of this report, detailed visuals for each risk are provided in the appendix. Readers are encouraged to consult *Appendix Figures 11–15* for full breakdowns by ACP (American Communities Project) type.

Key Findings:

- **Sea Level Rise:** Military Posts and Big Cities show elevated exposure, with 19% and 13% of counties respectively flagged as high risk. Coastal flooding threatens infrastructure and operational readiness.
- **Hurricanes:** The African American South faces the highest concentration of risk, with 64% of counties at red flag or high risk. Housing vulnerability and lack of insurance amplify recovery challenges.
- **Extreme Rainfall:** Middle Suburbs and Working Class Country are disproportionately affected, with over 50% of counties facing high flood risk. These areas often lack adaptive infrastructure.
- **Water Stress:** Hispanic Centers and LDS Enclaves—agriculture-dependent and rural—are projected to experience severe drought conditions, threatening food security and economic stability.
- **Heat Stress:** Evangelical Hubs and Rural Middle America show high exposure, with implications for labor productivity, public health, and energy demand.

These risks intersect with socioeconomic vulnerability, racial disparities, and infrastructure gaps. Many affected communities lack flood protection, cooling infrastructure, or access to adaptive financing. The NSS's emphasis on place-based

equity—via Justice40, CEJST targeting, and DOE Community Benefits Plans—aims to address these structural imbalances.

By embedding climate resilience into federal investment criteria, the NSS advances a typological model that is inducement-based, flexible, and regionally adaptive. However, implementation remains uneven, and cumulative risk exposure continues to outpace mitigation capacity in several regions.

Institutional Risks

- **Implementation capacity:** Complex interagency coordination and permitting processes may delay program delivery.
- **Political cycles:** Electoral changes could alter priorities, reduce continuity, or weaken long-term commitments.
- **Public trust:** Insufficient communication or perceived inequities in program benefits may erode legitimacy and support.

Mitigation Strategies

The NSS embeds risk mitigation through institutional, fiscal, and regulatory mechanisms:

- **Justice40 Initiative:** Directs 40% of federal climate and energy benefits to disadvantaged communities, improving resilience and equity
- **FERC Order 1920:** Mandates long-term transmission planning to address grid bottlenecks and interregional capacity gaps
- **DOE Community Benefits Plans:** Require applicants to quantify environmental, economic, and health outcomes in disadvantaged areas
- **State-level reforms:** Massachusetts' one-stop permitting law and other state initiatives accelerate infrastructure delivery

6.5. Strategic Recommendations

The evaluation of the 2022 Biden-Harris National Security Strategy (NSS) through classical economic planning typologies highlights both its strengths and its limitations. To enhance its role as a comprehensive national plan, several strategic recommendations are proposed.

1. Strengthen Intergovernmental Coordination

- Establish a National Inducement Strategy Council to harmonize subsidy programs across Commerce, Energy, Defense, and Treasury.
- Expand state strategic investment offices to maximize leverage of federal inducements and ensure regional equity.
- Encourage regional innovation compacts that pool local resources with federal incentives for critical sector development.

2. Enhance Long-Term Perspective Planning

- Develop a 15–25 year perspective plan nested within the NSS framework, providing continuity beyond electoral cycles.
- Integrate medium-term (5–10 year) rolling plans to bridge annual implementation with long-term objectives.
- Align perspective planning with climate targets, industrial transformation, and demographic trends.

3. Balance Physical and Financial Resource Mobilization

- Complement financial inducements with physical resource planning, ensuring adequate labor, land, water, and materials for infrastructure and energy projects.
- Strengthen statistical capacity to monitor resource availability and prevent bottlenecks.
- Introduce equity audits for programs exceeding \$10 billion to track geographic and demographic distribution of benefits.

4. Expand Public Engagement Mechanisms

- Pilot participatory budgeting for strategic infrastructure investments, allowing direct citizen input on discretionary funding.
- Establish regional civic engagement centers to facilitate community dialogue on NSS-aligned projects.

- Launch a National Strategy Transparency Dashboard to publicly track program outcomes, subsidy recipients, and equity metrics.

5. Build Resilience Against Uncertainties

- Institutionalize rolling reviews of major programs to recalibrate priorities in response to shocks.
- Strengthen strategic reserves and diversified supply chains to buffer against global disruptions.
- Expand equity safeguards to ensure disadvantaged communities benefit from climate and industrial transitions.

6.6. Principles of Good Planning

In the context of national development, *good planning* refers to the capacity of a country to design strategies that are both ambitious and achievable, forward-looking yet grounded in reality, inclusive of diverse societal needs, and responsive to change. Effective planning aligns national objectives with the lived experiences and requirements of citizens, ensuring that policies drive progress while remaining feasible to implement. It encourages innovation, promotes equitable growth, and coordinates development across sectors to prevent fragmentation. At the same time, good planning must remain flexible, enabling governments to recalibrate policies in response to unexpected challenges such as economic shocks, technological disruptions, or environmental crises. Strong institutional frameworks, coherent governance mechanisms, and alignment with global responsibilities further reinforce the quality and durability of national planning.

Within this framework, the 2022 U.S. National Security Strategy (NSS) demonstrates many of the hallmarks of good planning. Its approach to economic development, innovation, equity, environmental stewardship, and global engagement reflects a deliberate effort to balance ambition with practicality, national priorities with international commitments, and short-term resilience with long-term sustainability.

1. Meeting National Needs with Achievable Ambition

The NSS sets targets that are ambitious yet realistic, focusing on strengthening supply chains, advancing clean energy, and expanding workforce training. Rather than prescribing unattainable quotas, it balances feasibility with ambition by embedding programs such as the CHIPS and Science Act and the Inflation Reduction Act (IRA). These initiatives aim to raise productivity, enhance resilience, and stimulate investment across critical sectors, ensuring that development goals encourage progress without exceeding practical limits.

2. Stimulating Innovation and Modern Development Practices

Good planning requires the introduction of new ideas and tactics that drive efficiency and effectiveness. The NSS promotes innovation hubs, semiconductor clusters, and clean energy ecosystems designed to attract private investment and stimulate entrepreneurship. These initiatives introduce modern practices in manufacturing, energy storage, and carbon management, ensuring sustainable growth while addressing environmental concerns. By fostering technological advancement, the NSS positions the U.S. economy to remain competitive in a rapidly evolving global landscape.

3. Ensuring Comprehensive Sectoral Coverage

Fragmented development weakens national progress. The NSS avoids this risk by integrating agriculture, industry, services, infrastructure, environment, and governance under a unified framework. This comprehensive approach ensures that national priorities are pursued holistically, preventing gaps between economic, social, and environmental domains. Such integration reflects the principle that sustainable development requires coherence across all sectors of the economy.

4. Embedding Equity and Inclusiveness

Equity is central to good planning. Through the Justice40 initiative, the NSS directs 40% of climate and clean energy benefits to disadvantaged communities. This mechanism ensures that development is not only efficient but also fair, reducing disparities between regions and social groups. By embedding inclusiveness into national planning, the NSS strengthens social legitimacy and ensures that growth is broadly shared.

5. Maintaining Flexibility in the Face of Uncertainty

In a volatile global environment, flexibility is essential. The NSS incorporates rolling reviews and resilience planning, enabling recalibration in response to crises such as pandemics, supply chain disruptions, or climate extremes. This adaptive capacity ensures that national strategies remain relevant and effective even when confronted with unforeseen challenges. Flexibility, therefore, becomes a hallmark of resilience in modern economic planning.

6. Anchoring Planning in Institutional and Legislative Frameworks

Durable planning requires strong institutional foundations. The NSS is operationalized through major legislative acts — including the CHIPS Act, IRA, and Infrastructure Investment and Jobs Act (IIJA) — which provide fiscal mobilization and sectoral coverage. Institutional anchoring across federal agencies ensures coherence, accountability, and durability in implementation. This legislative foundation strengthens the credibility of the NSS and enhances its ability to deliver sustained outcomes.

7. Balancing National Priorities with Global Responsibilities

Finally, good planning requires alignment between domestic priorities and international responsibilities. The NSS situates U.S. policy within global frameworks such as the Paris Agreement, reinforcing climate leadership abroad while ensuring resilience at home. This dual orientation enhances credibility in international negotiations and strengthens domestic stability by embedding national development within a cooperative global order.

6.7. Methods of Planning

In national development, *methods of planning* refer to the strategic approaches governments employ to design, coordinate, and implement long-term objectives. These methods determine how national priorities are translated into actionable policies, how resources are mobilized and allocated, and how different actors—public agencies, private firms, and communities—participate in the development process.

In the United States, the National Security Strategy (NSS) integrates multiple planning methods to advance economic modernization, social equity, environmental protection, and global leadership in a coordinated and adaptive

manner. The following subsections examine five key planning methods embedded in the NSS—inducement planning, democratic planning, decentralized planning, functional planning, and flexible planning—highlighting how each contributes to a comprehensive and resilient national development framework.

1. Inducement Planning

Inducement planning relies on incentives rather than direct command. Within the NSS, this method is expressed through fiscal tools such as subsidies, tax credits, and public–private partnerships. Legislative acts like the CHIPS and Science Act, the Infrastructure Investment and Jobs Act (IIJA), and the Inflation Reduction Act (IRA) mobilize private investment in strategic sectors including semiconductors, clean energy, and advanced manufacturing.

This approach encourages private sector participation while aligning national priorities with long-term resilience. Inducement planning is particularly effective in dynamic environments where innovation and flexibility are required. However, its reliance on financial incentives can underemphasize the importance of physical resource allocation, such as land, minerals, and industrial capacity, which remain critical for sustainable development.

2. Democratic Planning

Democratic planning integrates governance with civic participation. The NSS is anchored in democratic institutions, legislative oversight, and stakeholder engagement. Programs such as Justice40 require agencies to consult disadvantaged communities, ensuring that benefits are equitably distributed.

This participatory approach strengthens legitimacy, transparency, and accountability, while balancing efficiency with fairness. Democratic planning reflects the U.S. commitment to inclusive growth and social justice. Yet, uneven state and local capacity can create variation in outcomes, highlighting the need for stronger institutional support at sub-national levels.

3. Decentralized Planning

Decentralized planning cascades national vision through multiple levels of government. While the NSS provides a federal strategic framework, implementation is delegated to states and local governments. This allows

flexibility in adapting programs to regional needs, such as technology hubs, workforce training, or climate resilience projects.

Decentralized planning ensures responsiveness and regional equity, empowering local actors to tailor strategies to their unique contexts. However, it also introduces risks of fragmentation if coordination mechanisms are weak. Effective decentralization therefore requires robust intergovernmental collaboration and clear accountability structures.

4. Functional Planning

Functional planning focuses on sectoral functions rather than rigid structural quotas. The NSS emphasizes functions such as energy transition, infrastructure modernization, supply chain resilience, and human capital development. By targeting these functions, the strategy avoids inflexible structural mandates and instead promotes adaptive sectoral strategies.

This method allows the U.S. to prioritize innovation ecosystems and climate resilience, ensuring that planning remains relevant to evolving challenges. However, functional planning may lack mechanisms for deep structural transformation, such as addressing inequality or modernizing outdated industrial bases. Without complementary structural reforms, functional planning risks reinforcing existing inefficiencies.

5. Flexible Planning

Flexible planning adapts to shocks and uncertainties. The NSS incorporates rolling reviews, resilience frameworks, and adaptive mechanisms to recalibrate priorities in response to crises such as pandemics, geopolitical disruptions, or climate extremes.

This flexibility is essential in a volatile global environment, ensuring that national security and economic development remain aligned. It reflects the principle that good planning must anticipate uncertainty and embed resilience. However, the absence of a formal 15–25 year perspective plan limits long-term foresight, leaving gaps in strategic vision for infrastructure transformation and climate adaptation.

6.8. Hierarchy of Planning

The *hierarchy of planning* provides a structured framework for organizing national priorities across different levels of governance and implementation. By moving from broad strategic vision to sectoral objectives and finally to specific projects, governments can solve problems more effectively, monitor progress, and ensure sustainability in national development. This layered approach allows coherence between long-term goals and short-term actions, ensuring that every initiative contributes to the overarching national vision.

1. Macro Planning

Macro planning represents the highest level of national planning. It establishes the strategic vision by analyzing current conditions alongside future trends, setting broad objectives that guide all subsequent levels of planning.

For the United States, the National Security Strategy (NSS) functions as macro planning. It defines overarching national priorities such as strengthening supply chains, advancing clean energy, promoting democratic governance, and ensuring resilience against crises. Importantly, the NSS integrates economic, social, environmental, and security dimensions under one umbrella, preventing fragmentation. It also positions the U.S. as a global leader in climate action and technological innovation, while embedding equity through initiatives like Justice40, which directs benefits to disadvantaged communities.

2. Sectoral Planning

Sectoral planning operates at the middle level, translating national vision into specific objectives for individual sectors. Each sector comprises agencies and institutions working within a defined domain, such as energy, industry, transportation, or human capital.

Within the NSS, sectoral planning includes:

- **Economic sector:** Revitalizing domestic manufacturing through the **CHIPS Act** and strengthening advanced technology hubs.
- **Energy and environmental sector:** Expanding renewable energy, modernizing grids, and committing to net-zero emissions by 2050.

- **Infrastructure sector:** Investing in transportation, broadband, and climate-resilient systems under the Infrastructure Investment and Jobs Act (IIJA).
- **Human capital sector:** Expanding workforce training, STEM pipelines, and apprenticeships to support industrial modernization.
- **Governance sector:** Reinforcing democratic institutions, transparency, and civic participation through Justice40 and interagency coordination.

Together, these sectors reduce inequality, enhance resilience, and promote balanced growth across regions. Sectoral planning ensures that national priorities are operationalized in ways that reflect the unique challenges and opportunities of each domain.

3. **Project Planning**

Project planning represents the most granular level of the hierarchy. It functions as the sub-unit of sectoral planning, focusing on specific initiatives that directly contribute to national development.

In the NSS, project planning includes:

- Establishing regional technology hubs to decentralize innovation capacity.
- Funding clean energy projects such as solar, wind, and carbon capture facilities.
- Implementing climate-resilient infrastructure projects like flood defenses, wildfire mitigation, and drought resilience systems.
- Supporting community-level programs under Justice40, ensuring disadvantaged areas benefit directly from federal investment.

These projects demonstrate how sectoral objectives are translated into tangible actions. They ensure that national strategies are not abstract visions but practical initiatives with measurable outcomes.

6.9. Linkage of Planning

The *linkage of planning* refers to the way different levels of the planning hierarchy interact to ensure coherence between national vision and local realities. In the National Security Strategy (NSS), this linkage is achieved through the combination of top-down and bottom-up approaches. By integrating both directions of planning, the NSS ensures that national security and development goals are not only strategically defined but also inclusively implemented.

1. Top-Down Planning

- Top-down planning begins with broad national objectives set at the federal level. The White House issues the NSS, which outlines macroeconomic, social, environmental, and security priorities. These overarching goals cascade downward to federal agencies, states, and local governments, providing a unified strategic direction.
- For example, the NSS sets national targets such as:
 - Achieving net-zero emissions by 2050
 - Strengthening supply chains
 - Revitalizing domestic manufacturing
- These macro goals are then translated into sectoral strategies through legislation such as the CHIPS and Science Act, the Infrastructure Investment and Jobs Act (IIJA), and the Inflation Reduction Act (IRA). At the operational level, agencies and sectors move into specific initiatives such as semiconductor production, clean energy projects, and workforce training programs. In this way, top-down planning provides the strategic overview first, ensuring that all subsequent actions align with national priorities.

2. Bottom-Up Planning

Bottom-up planning emphasizes input from local communities, stakeholders, and disadvantaged groups. Programs such as Justice40 require agencies to consult with communities to identify local needs and ensure that 40% of climate and clean energy benefits flow to disadvantaged populations.

This approach functions like assembling a puzzle: local feedback on issues such as energy access, environmental justice, or workforce training is aggregated into larger national strategies. For example, community-level projects in clean energy deployment, flood resilience, and broadband

expansion contribute directly to broader national targets of resilience and equity. Bottom-up planning ensures that national strategies remain grounded in lived realities, enhancing legitimacy and inclusiveness.

3. Combined Approach

The NSS employs both top-down and bottom-up planning simultaneously to maximize effectiveness. National targets such as climate resilience, technological leadership, and democratic governance are set at the macro level, while local projects and community engagement ensure that these goals are implemented inclusively and sustainably.

This combined approach strengthens coherence between federal vision and local realities. It ensures that national development is not only strategically coordinated but also socially responsive, making the NSS a multidimensional framework that balances ambition with inclusiveness, and vision with practicality.

6.10. Period of Planning

The *period of planning* provides the temporal framework through which national development strategies are organized. It ensures that immediate actions, medium-range reforms, and long-term transformations are aligned within a coherent structure. Within the National Security Strategy (NSS), planning is articulated across three horizons: short-term, medium-term, and long-term. Each reflects distinct objectives, instruments, and levels of ambition, together forming a multidimensional framework for national development.

1. Short-Term Planning

Short-term planning spans a few months to one year, focusing on urgent demands and immediate objectives. In the NSS, this is expressed through annual reviews, budget appropriations, and agency action plans.

- Examples include stabilizing supply chains, addressing inflationary pressures, and strengthening stockpiles of critical resources.
- These short-term actions provide the foundation for medium- and long-term strategies.

2. Medium-Term Planning

Medium-term planning covers more than one year but generally less than a decade. It addresses challenges requiring sustained effort over several years.

- In the NSS, medium-term planning is represented by multi-year legislative programs such as the CHIPS and Science Act (2022–2027), the Infrastructure Investment and Jobs Act (2022–2026), and the Inflation Reduction Act (2022–2030).
- These programs modernize infrastructure, expand clean energy, and strengthen advanced manufacturing.
- Medium-term planning also supports reforms in education, workforce training, and climate resilience, which require time to show tangible results.

3. Long-Term Planning

Long-term planning extends beyond a decade, addressing complex challenges that demand structural transformation.

- In the NSS, long-term commitments include achieving net-zero greenhouse gas emissions by 2050 and maintaining global leadership in technology and democratic governance.
- These goals require 20–25 years of sustained effort, involving innovation ecosystems, infrastructure modernization, and institutional reforms.
- Long-term planning ensures resilience against climate change, geopolitical shocks, and demographic shifts.

Comparative Table: Periods of Planning in the NSS

Period	Timeframe	Objectives	Key Instruments
Short-term	Months to 1 year	Immediate stabilization and crisis response	Annual reviews, budget appropriations, agency action plans
Medium-term	1–10 years	Sectoral modernization and resilience	Legislative programs
Long-term	10–25 years	Structural transformation and global leadership	Strategic commitments

6.11. U.S. Development Goals

The United States’ contemporary development agenda reflects a strategic effort to strengthen national competitiveness, enhance economic resilience, and advance equitable and sustainable growth. Guided by frameworks such as the National Security Strategy (NSS), the CHIPS and Science Act, and the Infrastructure Investment and Jobs Act (IIJA), U.S. priorities integrate economic modernization with social, environmental, and global objectives.

These priorities reaffirm the nation’s commitment to revitalizing domestic industries, upgrading critical infrastructure, and building a highly skilled workforce capable of supporting next-generation technologies. At the same time, the U.S. emphasizes social equity, climate action, and international cooperation as essential pillars of long-term stability and prosperity. Together, these goals illustrate a comprehensive national vision in which innovation, sustainability, and inclusiveness collectively drive future development.

1. Economic Growth

The U.S. seeks to strengthen domestic industries, support private sector innovation, and leverage advanced technologies to increase productivity and

create high-value jobs. Initiatives such as regional technology hubs and semiconductor manufacturing expansion under the CHIPS Act promote the digital economy. Export competitiveness is reinforced by securing supply chains and investing in clean energy, agriculture, and advanced manufacturing.

2. Industrialization and Modernization

Revitalization of U.S. manufacturing is pursued through high-tech industries such as semiconductors, renewable energy, and electric vehicles. The NSS emphasizes reshoring critical industries and transitioning to smart manufacturing systems for efficiency and resilience. Federal inducements encourage private investment in frontier technologies, positioning the U.S. as a leader in innovation ecosystems.

3. Infrastructure Development

Modernization of national infrastructure is advanced through the IIJA, which expands transportation networks, strengthens energy grids, and improves broadband connectivity. Projects include upgrading transmission systems to integrate renewable energy, expanding EV charging stations, and enhancing climate-resilient infrastructure to protect communities from floods, wildfires, and droughts.

4. Human Resource Development

Education and workforce training are expanded to align with labor market needs in advanced manufacturing, clean energy, and digital services. Programs emphasize STEM pipelines, apprenticeships, and community college partnerships. The NSS integrates human capital development with industrial modernization, ensuring that workers are equipped with the skills required for future competitiveness.

5. Social Equity

Fairness is embedded into national planning through initiatives like Justice40, which directs 40% of climate and clean energy benefits to disadvantaged communities. The NSS promotes equitable access to healthcare, education, and workforce opportunities, aiming to reduce poverty and regional disparities while strengthening democratic participation.

6. Environmental Protection

The NSS advances clean energy policies, enforces climate resilience measures, and invests in renewable energy technologies such as solar, wind, and carbon capture. Commitments include achieving net-zero emissions by 2050, with interim targets of a 50–52% reduction by 2030. Environmental protection is integrated into infrastructure modernization, ensuring sustainability and disaster resilience.

7. Global Integration

The U.S. reinforces leadership in global trade, climate governance, and technological alliances. The NSS balances domestic resilience with international commitments, including participation in the Paris Agreement and collaboration with allies on secure supply chains. By promoting democratic governance and climate finance, the U.S. strengthens its role as a global partner in sustainable development.

6.12 Structural Transformation and Equity Outcomes

While the 2022 Biden-Harris National Security Strategy (NSS) demonstrates strong functional planning through sectoral inducements and resilience frameworks, its capacity for structural transformation remains partial. A comprehensive national plan must not only mobilize investment but also reshape the underlying economic and social structures that determine long-term development outcomes. Three areas require closer integration:

1. Industrial Restructuring

The NSS advances reshoring initiatives through the CHIPS and Science Act and clean energy investments under the Inflation Reduction Act. These measures begin to modernize the U.S. industrial base, shifting production toward semiconductors, renewable energy, and advanced manufacturing. However, the strategy stops short of institutionalizing a broader restructuring agenda that addresses declining legacy industries, regional imbalances, and the transition from fossil fuels to sustainable energy systems. A more explicit structural framework would ensure that inducement policies translate into durable industrial transformation.

2. Inequality Reduction

Equity mechanisms such as Justice40 and regional technology hubs aim to distribute benefits more fairly, yet persistent disparities in income, wealth, and access to capital remain. Vulnerable communities often face barriers to workforce participation and clean energy adoption. To achieve structural equity, the NSS must embed redistributive measures—such as expanded education, workforce retraining, and targeted social investment—that go beyond fiscal inducements. These reforms would reduce systemic inequality and strengthen the legitimacy of national planning.

3. Labor–Capital Balance

The NSS emphasizes private-sector mobilization but underplays the structural challenges of workforce readiness. Shortages in skilled labor for semiconductors, clean energy, and advanced technologies highlight the need for coordinated labor market reforms. Expanding vocational training, STEM education, and apprenticeship programs would rebalance the labor–capital relationship, ensuring that inducement-based investment is matched by human capital capacity. Without such reforms, fiscal incentives risk being undermined by bottlenecks in labor supply.

Chapter 7

Conclusion & Recommendation

The evaluation of the 2022 Biden-Harris National Security Strategy (NSS) through classical economic planning typologies confirms that the NSS operates as a comprehensive national framework—effectively the United States’ closest equivalent to a national development plan. It integrates sectoral priorities, mobilizes fiscal inducements, reinforces democratic governance, and applies uniformly across states and regions. Its principal strengths lie in its inducement-based approach, democratic legitimacy, broad sectoral coverage, flexibility, and national scope. Together, these features position the NSS as a multidimensional framework that reconciles market efficiency with strategic state guidance.

At the same time, the analysis highlights structural gaps that limit the NSS’s durability as a long-term plan. The absence of a formal 15–25 year perspective reduces foresight in areas such as infrastructure transformation and climate adaptation. Heavy reliance on financial inducements without parallel physical resource planning risks underestimating constraints in labor, materials, and industrial capacity. Moreover, while the NSS incorporates adaptive mechanisms, it does not fully institutionalize rolling frameworks capable of anticipating unforeseen shocks. These limitations underscore the need for refinement if the NSS is to evolve into a truly sustainable national development strategy.

Recommendations for Strengthening the NSS

- **Institutionalize perspective planning** by embedding long-term (15–25 year) objectives within the NSS framework.
- **Balance financial and physical resource mobilization** to ensure adequate labor, materials, and infrastructure capacity.
- **Expand participatory mechanisms** to deepen public engagement and strengthen democratic legitimacy.
- **Enhance resilience** through rolling reviews, diversified supply chains, and adaptive governance structures.
- **Strengthen intergovernmental coordination**

Final Assessment

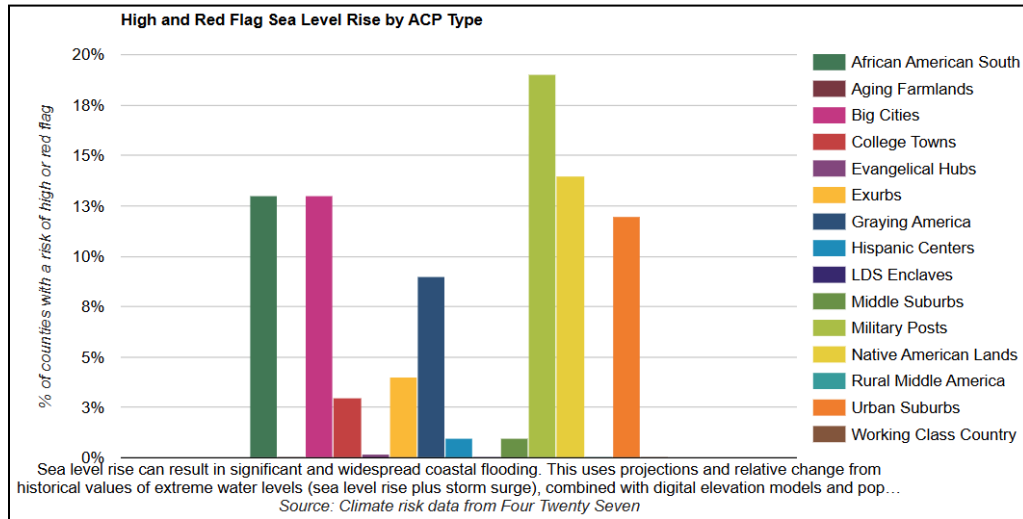
The 2022 Biden-Harris National Security Strategy (NSS) represents a multidimensional, inducement-driven, and democratic planning model that successfully reconciles market efficiency with strategic state guidance. By mobilizing fiscal incentives, embedding equity mechanisms, and coordinating across federal, state, and local levels, the NSS functions as the closest equivalent to a national development plan in the United States. Its strengths lie in its ability to integrate sectoral priorities, promote innovation, and balance domestic resilience with global leadership.

At the same time, the evaluation underscores the need for targeted adjustments to strengthen its coherence and durability. The absence of a formal long-term perspective plan spanning 15–25 years limits foresight in areas such as infrastructure transformation and climate adaptation. Heavy reliance on financial inducements without parallel physical resource mobilization risks underestimating constraints in labor, materials, and industrial capacity. Furthermore, while participatory mechanisms such as Justice40 enhance inclusiveness, deeper institutionalization of community engagement would reinforce democratic legitimacy and ensure equitable outcomes.

With these refinements—embedding perspective planning, balancing financial and physical resource mobilization, and expanding participatory governance—the NSS can evolve into a fully comprehensive national plan. Such a framework would be capable of sustaining resilience, equity, and competitiveness throughout the 2020s and beyond, positioning the United States to meet both domestic challenges and global responsibilities with strategic clarity and long-term vision.

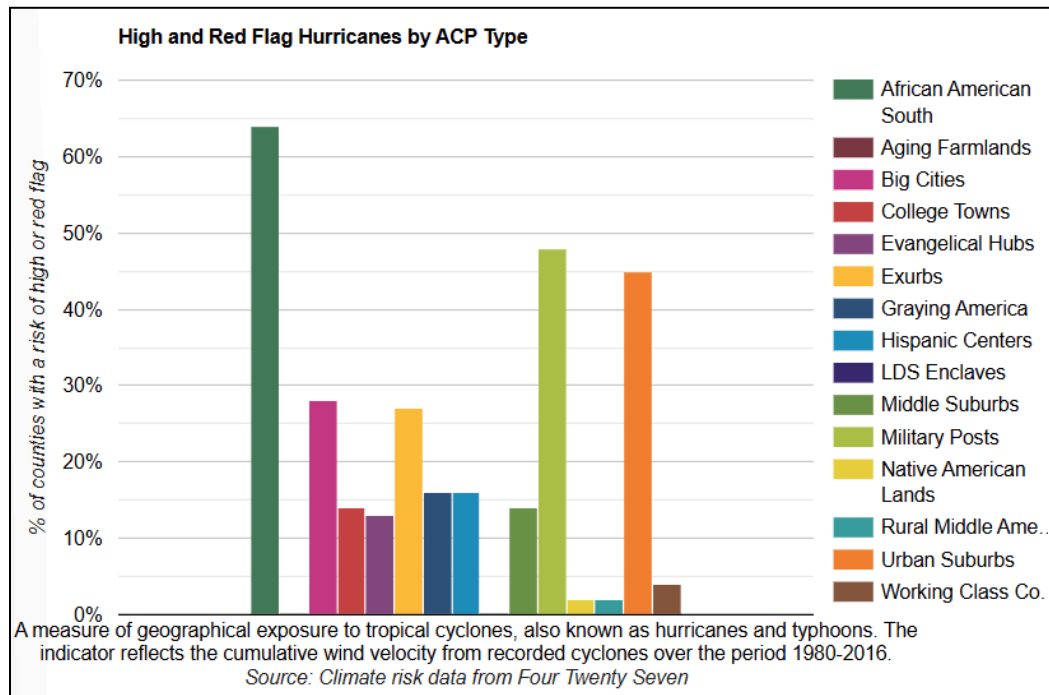
Appendix

Figure 11: High and Red Flag Sea Level Rise by ACP Type



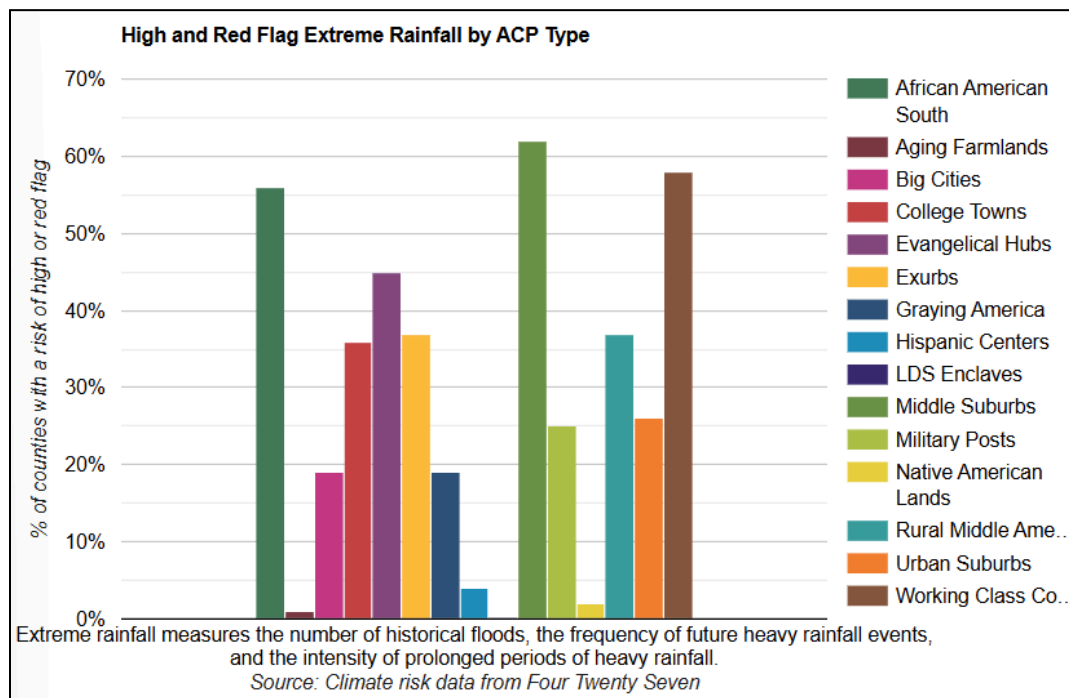
Source: Climate Risk Data

Figure 12: High and Red Flag Hurricane Risk by ACP Type



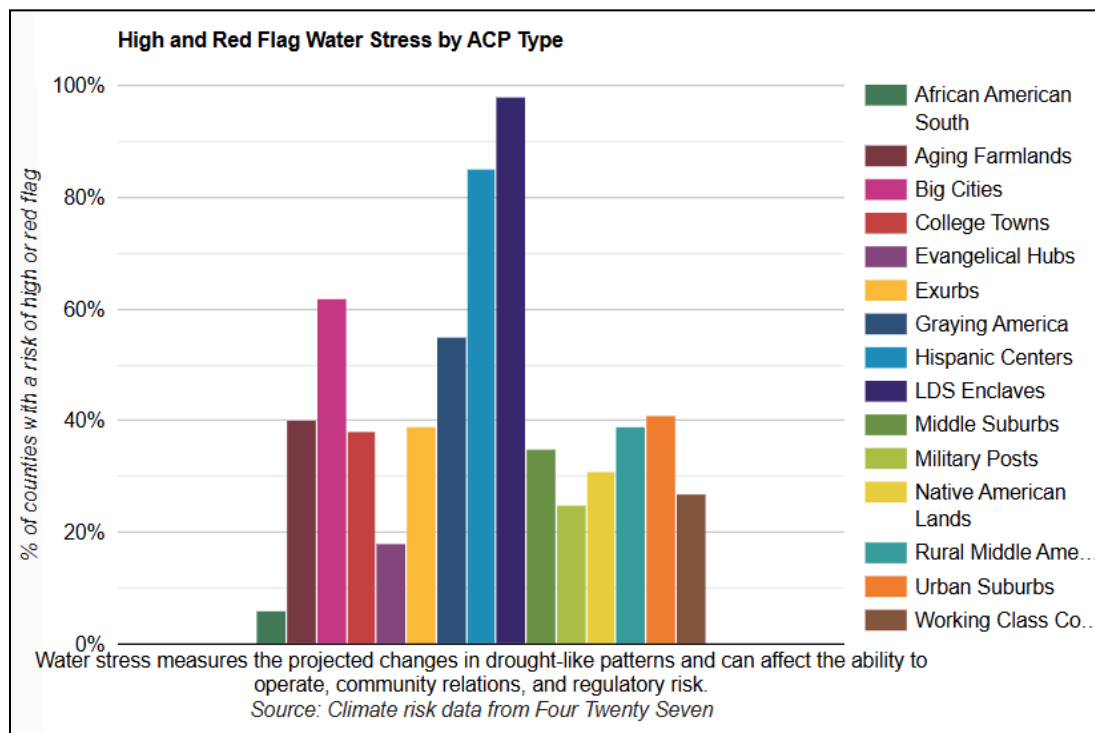
Source: Climate Risk Data

Figure 13: High and Red Flag Extreme Rainfall by ACP Type



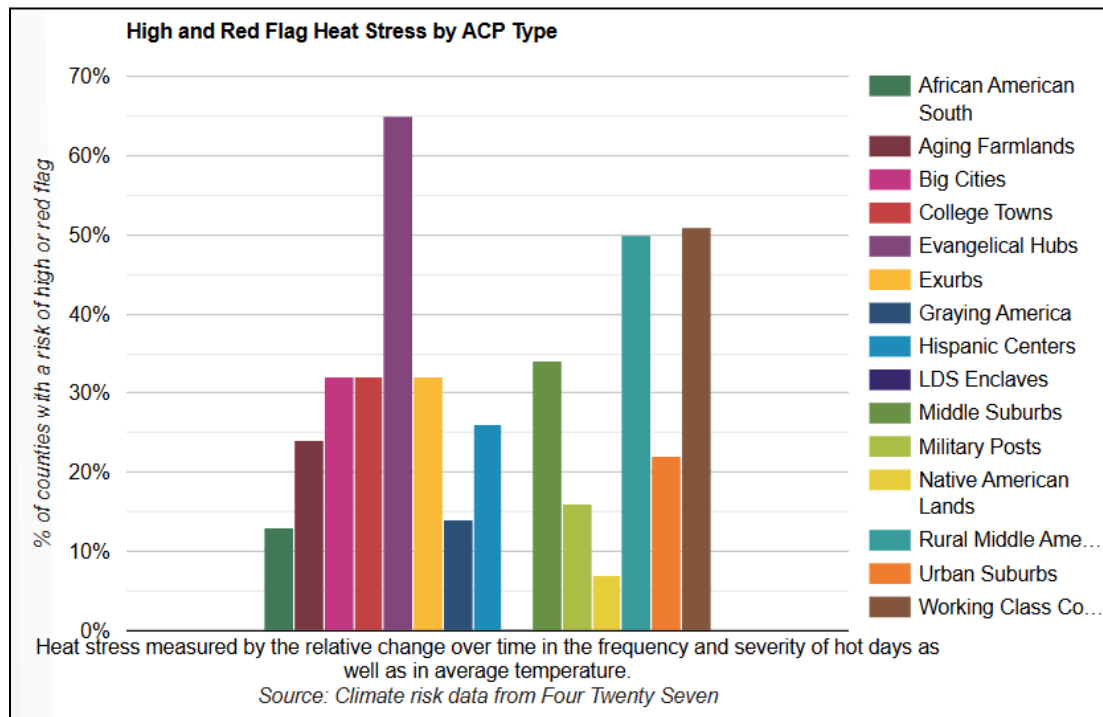
Source: Climate Risk Data

Figure 14: High and Red Flag Water Stress by ACP Type



Source: Climate Risk Data

Figure 15: High and Red Flag Heat Stress by ACP Type



Source: Climate Risk Data

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